

Uncertainty-Aware Transformer for Endometrioma Segmentation in Pelvic MRI

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Abstract—Automated segmentation of pelvic organs and endometriotic lesions in magnetic resonance imaging (MRI) enables quantitative disease assessment and surgical planning, yet remains challenging due to heterogeneous imaging protocols, small lesion size, and high inter-rater variability. The UT-EndoMRI dataset provides multi-sequence pelvic MRI from 132 subjects with expert annotations of uterus, ovaries, and endometriomas, revealing substantial labeling ambiguity and limited generalization of deterministic convolutional networks such as nnU-Net and RAovSeg. This project proposes an uncertainty-aware, multi-sequence Transformer framework based on a 3-D Swin-UNETR backbone. T1-, T2-, and fat-suppressed T1-weighted MRI sequences are fused through early convolutional harmonization, while uncertainty is modeled via Monte-Carlo dropout, test-time augmentation, and a probabilistic output head trained with a heteroscedastic loss. The model is pretrained on the homogeneous single-center Dataset 2 to learn robust organ features and fine-tuned on the multicenter Dataset 1 to specialize in endometrioma segmentation. Five-fold ensemble evaluation will report Dice, IoU, and HD95 for accuracy, and Expected Calibration Error, Brier Score, and uncertainty–error correlation for reliability. The approach aims to improve segmentation of endometriomas while providing voxel-wise confidence maps aligned with rater disagreement, supporting trustworthy deployment in clinical MRI analysis.

Index Terms—Endometriosis, Medical image segmentation, Pelvic MRI, Swin-UNETR, Transformer networks, Uncertainty quantification.

I. INTRODUCTION

Endometriosis is a chronic gynecological condition characterized by the presence of endometrial tissue outside the uterus, often leading to pain, infertility, and structural distortion of pelvic organs [1]. Magnetic resonance imaging (MRI) is the preferred modality for non-invasive assessment because it provides high soft-tissue contrast and allows visualization of endometriomas, deep infiltrating lesions, and pelvic organ displacement [2]. Accurate segmentation of pelvic structures and lesions from MRI can enable automated disease quantification, surgical planning, and longitudinal monitoring [3].

Despite advances in deep learning-based medical image analysis, segmentation of pelvic MRI in endometriosis remains difficult. Pelvic anatomy varies widely among patients, ovaries and endometriomas are small and irregular, and MRI sequences differ in contrast and acquisition parameters across scanners. Manual delineation by radiologists is labor-intensive

and subject to considerable inter-rater variability, particularly for ovaries and endometriomas. As a result, reliable automated segmentation remains an open challenge. The recently released UT-EndoMRI dataset reported Krippendorff’s $\alpha \approx 0.46$ for ovary segmentation, indicating high inter-rater variability [3].

Convolutional neural networks (CNNs), particularly U-Net [4] and its derivatives, have achieved strong results across many biomedical segmentation tasks. More recently, the self-configuring nnU-Net [5] extended this success through automated preprocessing and architecture adaptation, establishing a reliable baseline for medical imaging segmentation. In endometriosis MRI, nnU-Net and the customized RAovSeg pipeline [3] achieved reasonable performance on uterus and ovary segmentation but only modest Dice scores (≈ 0.29) for ovaries. These deterministic CNNs, however, offer no estimate of prediction confidence and can be over-confident on ambiguous boundaries. In heterogeneous pelvic MRI data, this lack of calibrated uncertainty undermines clinical reliability.

Transformers have emerged as powerful alternatives to CNNs for segmentation because their self-attention mechanism captures long-range dependencies and global anatomical context. Hybrid and fully Transformer-based architectures such as TransUNet [6], UNETR [7], and Swin-UNETR [8] have demonstrated improvements in multi-organ and brain-tumor segmentation tasks by combining self-attention with hierarchical feature learning. More recently, domain-specific foundation models such as RoMedFormer [9] have been introduced for 3-D genito-pelvic structure segmentation, further indicating the promise of Transformer architectures for female pelvic anatomy.

Although few studies have investigated Transformer models that explicitly quantify uncertainty, prior work on Bayesian SegNet [10], Monte-Carlo (MC) dropout [11], Deep Ensembles [12], and test-time augmentation (TTA) [13] established effective strategies for modeling epistemic and aleatoric uncertainty, while recent efforts [14], [15] have incorporated such mechanisms into Transformer backbones. Integrating these approaches can yield models that are both accurate and interpretable, which is crucial for clinical deployment in endometriosis imaging.

To address these gaps, this project investigates an uncertainty-aware Transformer-based model for multi-

sequence pelvic MRI segmentation in endometriosis. The goal is to leverage the global context modeling of Transformers together with probabilistic-uncertainty estimation to produce both accurate and interpretable segmentations. The framework will be developed using the UT-EndoMRI dataset [3], which provides multi-sequence MRI scans with expert annotations of uterus, ovaries, and endometriomas.

II. LITERATURE SURVEY

A. Datasets and Baselines

1) *UT-EndoMRI and RAovSeg* (Liang et al., 2025): Liang et al. released the UT-EndoMRI dataset, a multicenter, multi-rater pelvic MRI collection designed for endometriosis segmentation [3]. It includes 51 subjects with multi-rater manual annotations of uterus, ovaries, and visible endometriomas, and 81 single-rater cases for model development. The authors analyzed inter-rater reliability (Krippendorff’s $\alpha \approx 0.73$ for uterus, 0.46 for ovaries), demonstrating significant annotation ambiguity. They proposed a baseline pipeline, RAovSeg, combining a ResNet slice classifier and an attention U-Net segmenter, and compared it with nnU-Net. RAovSeg achieved Dice ≈ 0.29 for ovaries, slightly outperforming nnU-Net (≈ 0.27). Despite improvements, performance remained low, reflecting inherent difficulty and variability across scanners. The work established a reproducible benchmark for pelvic MRI segmentation and motivates methods that handle heterogeneity and quantify uncertainty, which are key objectives for the proposed uncertainty-aware Transformer model.

2) *nnU-Net* (Isensee et al., 2021): nnU-Net is a self-configuring deep-learning framework that automates all stages of medical image segmentation, from preprocessing to post-processing [5]. Instead of introducing a new architecture, it systematically adapts the U-Net design to each dataset through rule-based configuration derived from context-specific data “fingerprints.” The pipeline selects dimensionality (2-D/3-D), patch size, network depth, and hyper-parameters automatically, then ensembles multiple trained models. Tested on public benchmarks, nnU-Net matched or surpassed expert-engineered methods and became a de-facto standard for biomedical segmentation. Its deterministic outputs and strong generalization make it an ideal baseline for later uncertainty extensions such as MC dropout or TTA. In the UT-EndoMRI benchmark, nnU-Net’s moderate performance and lack of calibrated confidence underscore the need for architectures that capture dataset heterogeneity while providing reliable uncertainty estimation—objectives pursued in this project.

3) *From UNet to Res-UNet and nnU-Net* (Huang et al., 2024): Huang et al. conducted a controlled comparison of four CNN-based segmentation architectures—UNet, Res-UNet, Attention Res-UNet, and nnU-Net—across three medical imaging tasks: brain tumor MRI, polyp detection, and cardiac MRI [16]. Each model was trained under identical hyper-parameters, and results were evaluated by Dice, IoU, precision, and recall. Res-UNet achieved the best Dice (0.863) and IoU (0.759) on brain tumors, while nnU-Net produced the highest recall (0.881) and accuracy (0.994). For polyp segmentation,

nnU-Net clearly outperformed all others (Dice ≈ 0.94). The study confirmed nnU-Net’s robustness across modalities but also revealed limitations of purely convolutional architectures in capturing long-range dependencies. The authors recommended exploring Transformer-based or 3-D extensions, reinforcing the motivation for this project’s uncertainty-aware Transformer approach.

B. Transformer-Based Segmentation

1) *TransUNet* (Chen et al., 2021): TransUNet introduced a hybrid encoder–decoder framework that merges CNN feature extraction with a Vision Transformer (ViT) encoder and U-Net-style skip connections [6]. CNN layers first capture low-level spatial details, whose feature maps are tokenized and fed into Transformer blocks to model global dependencies. A cascaded upsampling decoder reconstructs dense predictions while re-injecting high-resolution CNN features. Evaluated on Synapse multi-organ CT and ACDC cardiac MRI datasets, TransUNet achieved Dice $\approx 77.5\%$ and 89.7% , outperforming U-Net and other CNN baselines. The work demonstrated that combining self-attention with convolution improves contextual understanding and boundary precision. This architecture pioneered the hybrid design later refined in UNETR and Swin-UNETR, providing the conceptual foundation for the Transformer backbone adopted in this project.

2) *UNETR* (Hatamizadeh et al., 2021): The UNETR architecture reformulates 3-D medical image segmentation as a sequence-to-sequence task by using a pure Transformer encoder connected to a CNN decoder through multi-scale skip connections [7]. Volumetric inputs are divided into non-overlapping 3-D patches, linearly embedded, and processed by stacked multi-head self-attention layers to capture global contextual information. Decoder stages reconstruct the segmentation via deconvolutions using intermediate Transformer representations. Evaluated on BTCV multi-organ CT and Medical Segmentation Decathlon tasks, UNETR achieved state-of-the-art Dice scores (~ 0.90 BTCV) with notable gains on small organs compared with 3-D U-Net and V-Net. The design efficiently balances global context and local detail while keeping computation moderate. By operating directly on volumetric patches, UNETR demonstrates the feasibility of full-Transformer modeling in 3-D data and motivates using similar architectures for pelvic MRI, where spatial continuity across slices is critical.

3) *Swin-UNETR* (Hatamizadeh et al., 2022): Swin-UNETR replaces the UNETR encoder with a hierarchical 3-D Swin Transformer employing shifted-window self-attention [8]. The windowing mechanism limits attention cost to $\mathcal{O}(n)$ per window while enabling inter-window connections via periodic shifts, thereby preserving global consistency with lower computational burden. Feature maps from each encoder stage are merged with CNN decoder blocks through U-Net-style skip connections to recover fine details. Trained with Dice + Cross-Entropy loss on BTCV and MSD datasets, Swin-UNETR surpassed both CNN baselines and pure ViT encoders in Dice and Hausdorff-95 metrics, particularly for multi-organ and

small-structure segmentation. The method demonstrated faster convergence and fewer parameters than UNETR while retaining global context modeling. Its efficiency and modularity make Swin-UNETR an appealing backbone for this project’s uncertainty-aware model, especially when GPU memory is constrained and multi-sequence fusion is required.

4) *RoMedFormer* (Zhang et al., 2025): RoMedFormer is a 3-D Transformer foundation model targeting segmentation of female genito-pelvic structures in MRI and CT [9]. The framework uses rotary positional embeddings to encode 3-D spatial relationships more effectively than absolute embeddings. Training follows a three-stage pipeline: self-supervised masked-volume pre-training on large unlabeled CT data, supervised multi-organ fine-tuning on public datasets, and task-specific adaptation to small institutional MRI/CT pelvic cohorts. On test volumes it achieved average Dice ≈ 0.71 , with > 0.86 for certain organs such as the bulboclititoris. The model highlights how large-scale pre-training and modality-adaptive Transformers can generalize to under-represented anatomical domains. For this work, RoMedFormer serves as a domain-specific reference showing the potential of Transformer foundations in female pelvic anatomy and motivating further exploration of uncertainty and multi-sequence integration for endometriosis segmentation.

5) *Self-Supervised Pre-Training for Swin Transformers* (Tang et al., 2024): This study extends Swin-UNETR by introducing self-supervised pre-training tasks involving masked-volume inpainting, 3-D rotation prediction, and contrastive representation learning to exploit large unlabeled medical datasets [17]. Pre-training on 5,050 CT volumes yielded richer volumetric representations that improved fine-tuning Dice scores by $\approx 2\%$ on BTCV and $\approx 10\%$ when labeled data were reduced to 10%. The approach substantially reduced annotation requirements while maintaining state-of-the-art segmentation accuracy. Results confirmed that Transformers benefit strongly from self-supervised objectives, particularly in low-label scenarios. For the present endometriosis project, these findings justify incorporating pre-trained Transformer weights or domain-specific self-supervision on unlabeled pelvic MRIs before uncertainty-aware fine-tuning, improving robustness under the limited-data conditions typical of UT-EndoMRI.

C. Uncertainty Modeling and Quantification

1) *MC Dropout as Bayesian Approximation* (Gal & Ghahramani, 2016): Gal and Ghahramani showed that applying dropout at train and test time yields a variational approximation to a Bayesian deep Gaussian process, enabling epistemic uncertainty estimation without architectural changes [11]. At inference, multiple stochastic forward passes (MC dropout) provide a predictive mean and variance; the variance correlates with model ignorance and typically rises out of distribution. Across regression and classification benchmarks, MC dropout improved negative log-likelihood and yielded better uncertainty–error alignment than point-estimate networks, with modest compute overhead proportional to the number of samples. For medical image segmentation, MC

dropout can be placed in encoder, bottleneck, and decoder blocks to produce voxel-wise uncertainty maps. This makes it a practical first mechanism for the proposed uncertainty-aware Transformer: insert dropout in attention and MLP sublayers, sample $T \approx 8\text{--}16$ times at test time, and aggregate per-voxel entropy/variance to flag low-confidence regions in pelvic MRI.

2) *Bayesian SegNet* (Kendall et al., 2017): Kendall et al. extended encoder–decoder segmentation (SegNet) with MC dropout at test time to obtain pixel-level posterior uncertainty while improving mean IoU by $\sim 2\text{--}3\%$ over deterministic baselines on CamVid, SUN RGB-D, and PASCAL VOC [10]. The study analyzed where to place dropout (encoder, decoder, or both), showing that uncertainty peaks at class boundaries, rare classes, and occlusions—locations that often coincide with segmentation errors. Crucially, uncertainty maps were actionable: they identified regions likely to benefit from human review or active learning. This paper is an early template for turning dense prediction networks into probabilistic segmenters with minimal code changes. For endometriosis MRI, the same principle applies to Transformer decoders: maintain dropout during inference to quantify epistemic uncertainty, visualize per-voxel entropy over ovaries/endometriomas, and correlate uncertainty with inter-rater disagreement reported in UT-EndoMRI.

3) *Deep Ensembles* (Lakshminarayanan et al., 2017): Deep Ensembles estimate predictive uncertainty by averaging outputs from independently initialized and trained networks optimized with proper scoring rules [12]. The method scales trivially, is architecture-agnostic, and consistently improves calibration (negative log-likelihood, Brier score) and robustness under distribution shift versus single models and MC dropout on UCI regression, MNIST/SVHN, and ImageNet. Ensembles capture a broader slice of epistemic uncertainty through diversity in weight space, and accuracy typically rises with ensemble size M . The trade-off is linear inference cost and memory with M , but parallelism and test-time model selection can mitigate latency. For this project, ensembling 3–5 Swin-UNETR backbones over cross-validation folds yields strong mean masks and reliable uncertainty (variance across members). Ensemble variance can be fused with aleatoric measures to produce composite uncertainty maps guiding radiologist attention in ambiguous pelvic regions.

4) *Aleatoric Uncertainty with Test-Time Augmentation* (Wang et al., 2019): Wang et al. formalized TTA as sampling from an image-acquisition model to measure aleatoric uncertainty in segmentation [13]. By applying stochastic geometric/intensity transforms to each test image and aggregating predictions, they improved Dice and reduced ASSD on fetal brain and BraTS tumor segmentation versus single-pass baselines and simple test-time dropout. Per-voxel entropy across TTA samples correlated with segmentation errors better than MC dropout alone, indicating complementary uncertainty sources: input-driven (aleatoric) versus parameter-driven (epistemic). The hybrid TTA+MC approach offered the best error localization at modest extra compute. For pelvic MRI, TTA captures variability from positioning, scale, and intensity differences

across scanners/sequences. Combined with MC dropout or ensembles in a Transformer backbone, it provides calibrated voxel-wise confidence maps that align with the UT-EndoMRI multi-rater variability, supporting safer clinical interpretation.

5) *Evaluating Uncertainty Quantification in Medical Segmentation (MDPI, 2024)*: A large multi-dataset, multi-algorithm study compared MC dropout, ensemble, and regression models across knee MRI, sacroiliac joint MRI, lung CT, and chest X-ray segmentation datasets [13]. The authors introduced the Gray-level Dice Coefficient (GDC) metric for evaluating “soft” predictions. They found that changing dropout rate (20–80%) and adding multiple annotators had limited systematic effect on uncertainty calibration—benefit depends on dataset complexity. Ensemble and MC-dropout models performed similarly, while regression networks underperformed. The work highlights that uncertainty methods must be evaluated per dataset and that multiple metrics beyond Dice (e.g., normalized root-mean-square error, normalized mutual information, GDC) are needed. For this project, the paper provides an important benchmarking practice: evaluate both segmentation accuracy and uncertainty calibration on UT-EndoMRI and correlate uncertainty with inter-rater variability rather than relying solely on Dice.

D. Uncertainty-Aware Transformers

1) *Uncertainty-Aware Transformer via Mean Teacher (Wang et al., 2022)*: Wang et al. proposed a semi-supervised Vision Transformer trained with a mean-teacher framework for cardiac MRI segmentation [14]. The student network learns from labeled data using supervised loss and from unlabeled data via a consistency loss against the teacher (an exponential moving average of the student). Pixel-wise uncertainty from MC dropout filters out unreliable pseudo-labels during training, gradually including more uncertain regions as training stabilizes. On the MICCAI 2017 cardiac dataset, the method outperformed U-Net-based and adversarial semi-supervised baselines even with $< 10\%$ labeled data. This demonstrates that embedding uncertainty directly into training rather than post hoc stabilizes learning and improves generalization. This paper motivates integrating uncertainty filtering or consistency regularization within a Transformer backbone to leverage unlabeled slices from UT-EndoMRI’s multi-sequence data.

2) *Uncertainty-Aware Adapter for SAM (Chen et al., 2024)*: Jiang et al. introduced UA-SAM, an adapter-based extension of the Segment Anything Model that generates multiple plausible segmentations to reflect annotation ambiguity [15]. The adapter includes a conditional variational autoencoder producing latent uncertainty samples, and a Condition-Modified Sample Module that injects those samples into SAM’s encoder blocks. At inference, multiple latent draws yield diverse segmentation hypotheses whose variance represents uncertainty. On LIDC-IDRI (lung CT) and REFUGE2 (fundus) datasets, UA-SAM achieved Dice ≈ 0.89 and 0.86 , outperforming deterministic SAM and prior uncertainty models. The work demonstrates that uncertainty can be added to large pretrained segmenters through lightweight adapters without retraining

the backbone. For endometriosis MRI, it suggests a practical route: combine pretrained medical Transformers with small probabilistic adapters to capture annotation ambiguity and produce uncertainty-weighted pelvic segmentations.

III. PROBLEM STATEMENT

Accurate segmentation of pelvic organs and endometriotic lesions in MRI is essential for automated assessment, surgical planning, and disease progression monitoring in endometriosis [3]. However, the task remains challenging due to several factors: (1) highly variable pelvic anatomy across patients, (2) small and irregular shapes of ovaries and endometriomas, (3) heterogeneous MRI acquisition protocols between scanners and sequences (T1-weighted, T2-weighted, and fat-suppressed T1), and (4) limited availability of high-quality, consistently annotated training data. Even among expert radiologists, inter-rater variability in ovary and lesion boundaries is substantial, indicating intrinsic annotation ambiguity [3].

The recently released UT-EndoMRI dataset [3] established baselines using convolutional architectures such as nnU-Net [5] and RAovSeg, which achieve reasonable performance for uterus and ovary segmentation but only modest Dice scores (≈ 0.29) for ovaries and exhibited high inter-rater variability (Krippendorff’s $\alpha \approx 0.46$). These findings highlight two unresolved issues:

- 1) Limited generalization and reliability of deterministic CNN-based models in heterogeneous pelvic MRI data, where organ shape, contrast, and imaging protocols vary widely [3], [5].
- 2) Absence of calibrated uncertainty quantification as current models output single, deterministic masks without indicating prediction confidence or inter-rater ambiguity [10]–[15].

Such limitations hinder clinical deployment because radiologists require both segmentation accuracy and awareness of confidence in ambiguous regions (e.g., endometriomas partially obscured by bowel gas or variable ovarian signal). Therefore, a new segmentation framework is needed that can (a) capture long-range spatial dependencies within the pelvis, (b) operate robustly under limited annotated data, and (c) quantify both epistemic and aleatoric uncertainty to guide clinical interpretation.

This project aims to develop an uncertainty-aware Transformer-based segmentation model for multi-sequence pelvic MRI in endometriosis. The proposed model will integrate a Vision Transformer backbone [6]–[8] with uncertainty estimation mechanisms such as MC dropout [11], TTA [13], or probabilistic output layers to jointly predict segmentation masks and voxel-wise confidence maps. By explicitly modeling epistemic and aleatoric uncertainty [11], [13], the framework will identify low-confidence regions corresponding to ambiguous or poorly defined boundaries. The approach is expected to improve model robustness under small data regimes and provide interpretable outputs that align with inter-rater variability, thereby enhancing trust and clinical applicability in automated pelvic MRI segmentation. The model

will also investigate multi-sequence fusion of T1-, T2-, and fat-suppressed T1-weighted MRI to exploit complementary anatomical and lesion-contrast information [3].

IV. DATASET DESCRIPTION

This project uses the UT-EndoMRI dataset [3], a publicly released multicenter pelvic MRI resource for deep-learning analysis of endometriosis. The collection contains 132 subjects across two cohorts and is distributed in NIfTI format with acquisition metadata (Zenodo DOI 10.5281/zenodo.15750762). Each case includes T1-weighted (T1w), T2-weighted (T2w), and fat-suppressed T1 (T1-FS) sequences, with voxel-wise labels for uterus, ovaries, and endometriomas where visible. The dataset was approved for research use under institutional ethics protocols and provides the first benchmark for automated segmentation in endometriosis MRI.

Two cohorts are included. Dataset 1 comprises 51 subjects collected from 15 institutions across nine scanner models (Siemens, GE, Philips) at 1.5 T and 3 T field strengths. Each case was independently annotated by one to three radiologists to assess inter-rater agreement. Dataset 2 contains 81 subjects acquired on a single Philips Ingenia 1.5 T scanner with standardized protocols and a single expert rater, used to develop automated ovary-segmentation pipelines. Every subject has uterus and ovary annotations; endometrioma masks exist only when lesions are radiologically detectable.

TABLE I
SUMMARY TABLE

Dataset	Inst. Count	Subjects	Raters/Case	MRI Sequences	Field Strengths
D1_MHS	15	51	1–3	T1w, T2w, T1-FS	1.5 T, 3 T
D2_TCPW	1	81	1	T1w, T2w, T1-FS (+ T2-FS)	1.5 T

Annotations were drawn in 3D Slicer following a standardized protocol and reviewed by senior radiologists. Inter-rater agreement analysis on Dataset 1 yielded Krippendorff’s $\alpha = 0.73 \pm 0.12$ for uterus and 0.46 ± 0.19 for ovaries, confirming notable ambiguity. Mean pairwise Dice similarity was 0.73 ± 0.18 (uterus) and 0.48 ± 0.24 (ovaries). Such variability establishes the dataset as an ideal test bed for uncertainty-aware segmentation.

Exploratory analysis aggregated sequence and label information across both cohorts. Typical D2_TCPW volumes measure $352 \times 352 \times 36$ voxels with spacing $\approx 0.68 \times 0.68 \times 6$ mm and raw intensity range $[0, 3181]$. Mean physical volumes were uterus ≈ 200.8 cc, ovary ≈ 19.8 cc, and endometrioma ≈ 86.1 cc, exhibiting strong inter-subject variability (uterus $\sigma \approx 221.8$ cc).

Figure 1 shows structure-wise intensity distributions for a representative T2-FS volume. Cyst voxels exhibit the highest mean intensity ($\mu \approx 1681$, $\sigma \approx 266$), ovaries intermediate ($\mu \approx 1489$, $\sigma \approx 506$), and uterus lowest ($\mu \approx 646$, $\sigma \approx 321$).

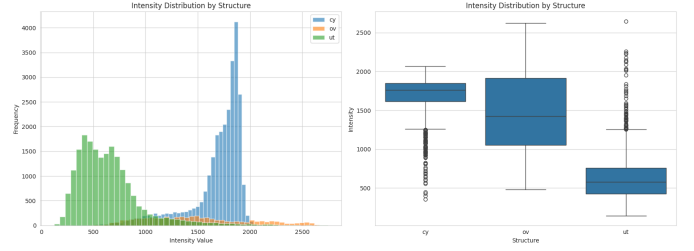


Fig. 1. Intensity distributions within labeled pelvic structures for a representative T2-FS scan. Left: histogram of voxel intensities for cyst (cy), ovary (ov), and uterus (ut). Right: corresponding boxplots showing interquartile ranges and outliers. Cyst regions show higher mean intensities than ovary and uterus, illustrating contrast differences exploitable for sequence-aware fusion.

The accompanying boxplots demonstrate substantial overlap yet clear separation of median intensities, supporting the use of sequence-specific normalization and fusion to exploit contrast differences among structures. Foreground voxels represented $< 2\%$ of the total volume (background:foreground $\approx 85 : 1$), motivating class-balanced sampling, Dice + Cross-Entropy or Tversky/Focal losses, and uncertainty calibration to highlight low-support regions.

V. MODEL DESCRIPTION

The proposed network extends the Swin-UNETR backbone [8] to incorporate multi-sequence fusion and uncertainty estimation. Swin-UNETR employs a 3-D hierarchical Transformer encoder with shifted-window self-attention and a convolutional decoder linked through skip connections. For this project, the encoder receives fused inputs from the T1w, T2w, and T1-FS volumes. Fusion occurs by channel concatenation followed by a shallow $1 \times 1 \times 1$ convolution to equalize dynamic ranges across sequences before patch embedding. This hybrid design preserves CNN-style locality in early layers while modeling long-range dependencies through multi-head self-attention, which is essential for delineating spatially separated pelvic organs and endometriotic lesions.

Because endometrioma masks are concentrated in the multicenter Dataset 1 (D1_MHS), the model will be pretrained on the homogeneous single-center Dataset 2 (D2_TCPW) to learn robust uterus and ovary representations, then fine-tuned on endometrioma-positive subjects from D1. Soft consensus masks (majority vote of available raters) will be used for training, while a held-out subset of D1 will be reserved for multi-rater testing and uncertainty analysis. This sequential training scheme leverages D2’s consistency for initialization and D1’s diversity for lesion learning and generalization. We integrate three complementary mechanisms:

- MC Dropout:** Dropout layers remain active during inference in the Transformer’s MLP and attention sub-blocks, enabling T stochastic passes per model [11].
- Test-Time Augmentation (TTA):** Each test volume undergoes random flips and small affine perturbations, yielding K augmented predictions per pass [13].
- Probabilistic Output Head (optional):** The decoder predicts per-voxel class logits with mean μ and log-

scale $\log \sigma$; see the training objective below, inspired by variational approximations [11].

At inference we use M independently trained models (5-fold CV), T MC-dropout passes, and K TTA samples. Let $p_c^{(m,t,k)}(x)$ denote the softmax probability for class c at voxel x from model m , MC pass t , and TTA sample k . The ensemble-TTA-MC mean is, following [11]–[13],

$$\bar{p}_c(x) = \frac{1}{MTK} \sum_{m=1}^M \sum_{t=1}^T \sum_{k=1}^K p_c^{(m,t,k)}(x). \quad (1)$$

The predicted mask is $\hat{y}(x) = \arg \max_c \bar{p}_c(x)$. We report:

$$H(x) = -\frac{1}{\log C} \sum_{c=1}^C \bar{p}_c(x) \log \bar{p}_c(x), \quad (2)$$

$$V_{\text{epi}}(x) = \frac{1}{C} \sum_{c=1}^C \text{Var}_{m,t}[\mathbb{E}_{h_{avek}}[p_c^{(m,t,k)}(x)]], \quad (3)$$

$$V_{\text{ale}}(x) = \frac{1}{C} \sum_{c=1}^C \mathbb{E}_{m,t}[\text{Var}_k[p_c^{(m,t,k)}(x)]], \quad (4)$$

where C is the number of classes.

Predictive entropy (2) follows [10], [13]; epistemic variance (3) follows variation across models/MC [11], [12]; aleatoric variance (4) follows variation over TTA [13].

If a composite is required, weights will be selected on a validation split.

The baseline objective is

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{Dice}} + \lambda_1 \mathcal{L}_{\text{CE}}, \quad (5)$$

which is standard in Swin-UNETR/UNETR-based segmentation [7], [8]. With the optional probabilistic head, logits are sampled as

$$z(x) = \mu(x) + \sigma(x) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (6)$$

and we optimize the Monte-Carlo expectation of cross-entropy with a sparsity prior on the scale, following the variational treatment of stochastic logits [11]:

$$\mathcal{L}_{\text{prob}} = \mathcal{L}_{\text{Dice}} + \lambda_1 \frac{1}{S} \sum_{s=1}^S \mathcal{L}_{\text{CE}}(\text{softmax}(z^{(s)}), y) + \lambda_2 \|\log \sigma\|_1. \quad (7)$$

Setting $\lambda_2=0$ and $S=1$ recovers $\mathcal{L}_{\text{base}}$.

AdamW (initial learning rate 1×10^{-4} , $\beta_1=0.9$, $\beta_2=0.999$) with cosine decay and 10-epoch linear warm-up. Training runs for 300 epochs with AMP on 24 GB GPUs. Pretraining is performed on D2_TCPW, followed by fine-tuning on D1_MHS.

Five independently initialized models are trained via five-fold cross-validation. Performance will be reported using Dice Similarity Coefficient (DSC), Intersection-over-Union (IoU), and Hausdorff Distance (HD95) for accuracy, alongside uncertainty–error correlation for calibration. All models will be pretrained on D2 and fine-tuned on D1 to target endometrioma segmentation, while a held-out D1 split will serve for final evaluation. These experiments will assess how uncertainty modeling and multi-sequence fusion influence segmentation

reliability and robustness in heterogeneous pelvic MRI. Ablations will vary (i) fusion strategy (early vs. cross-attention), (ii) uncertainty configuration (none / MC / MC+TTA / probabilistic head), and (iii) ensemble size.

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